

Examples of Objects Detection Data sets

VOC 2007 / 2012:

- 20 classes
- e.g. person, cat, dog, car, chair, bottle

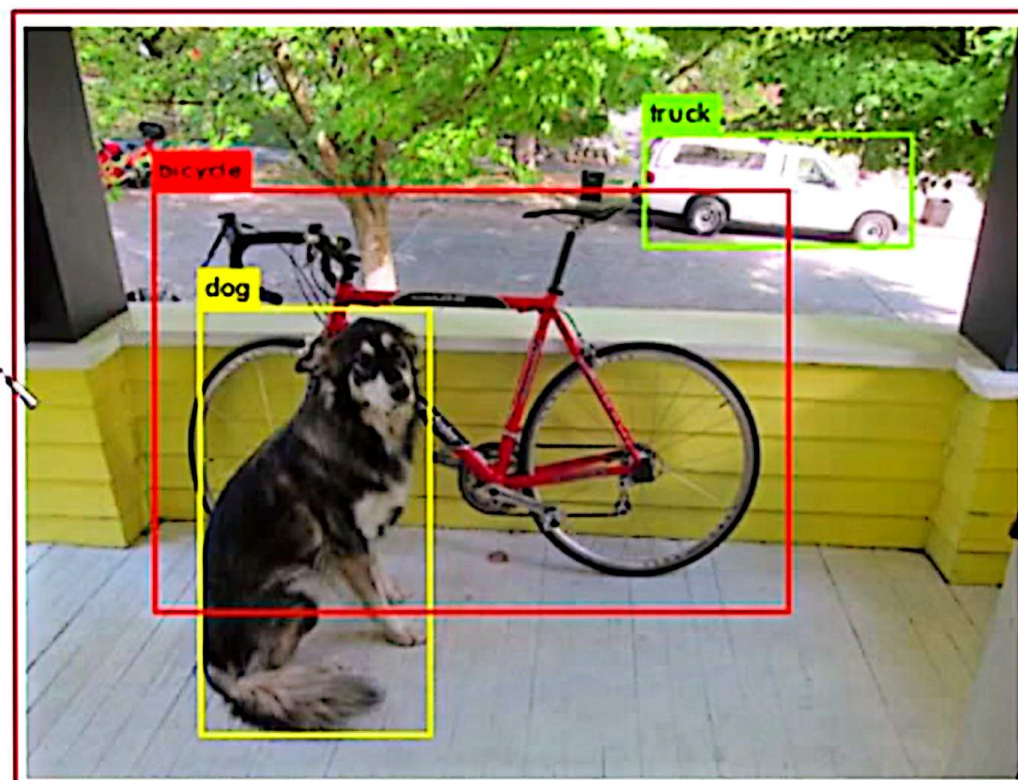
MS COCO:

- 80 classes
- e.g. book, apple, teddy bear, scissors

ImageNet1000:

- 1000 classes
- e.g. German shepherd, golden retriever, European fire salamander

YOLO – The Detected Output



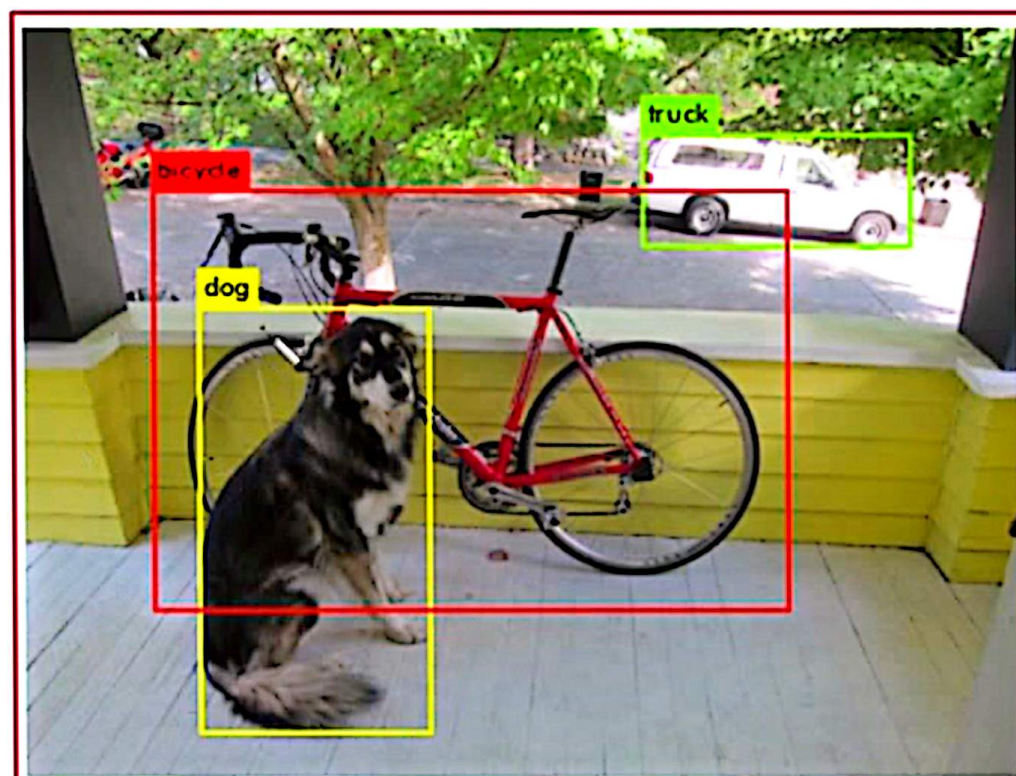
Confidence
of Classes {

```
Loading weights from yolov3.weights...Done!  
data/dog.jpg: Predicted in 21.447420 seconds.  
dog: 100%  
truck: 92%  
bicycle: 99%
```

YOLO – The Input Image



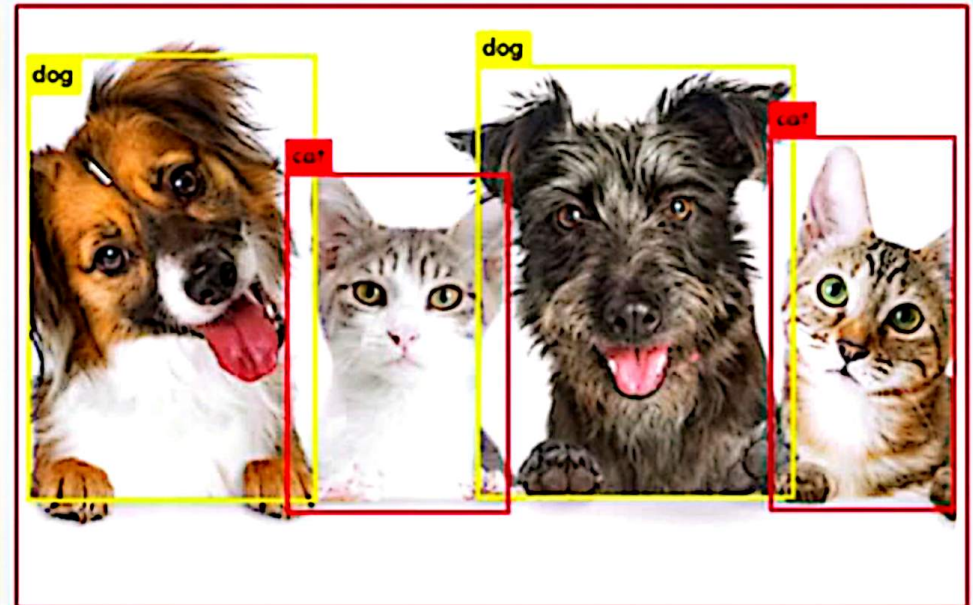
YOLO – The Detected Output



Confidence
of Classes {

```
Loading weights from yolov3.weights...Done!  
data/dog.jpg: Predicted in 21.447420 seconds.  
dog: 100%  
truck: 92%  
bicycle: 99%
```

YOLO – The Detected Output



YOLO – The Input Image

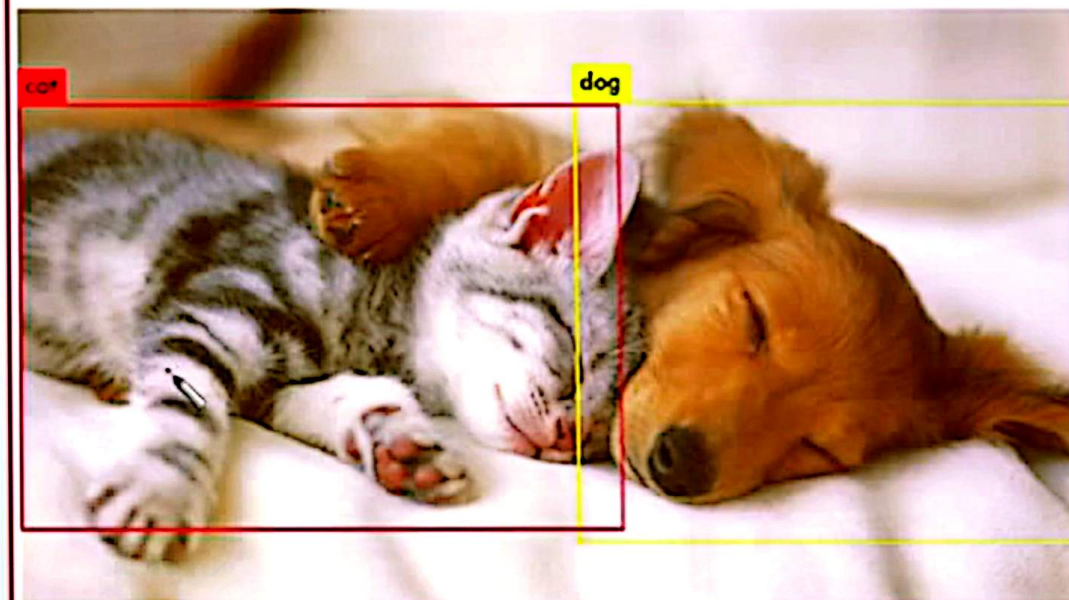


Currently, it has just been saved...

4:15 / 1:04:08

0

YOLO – The Detected Output



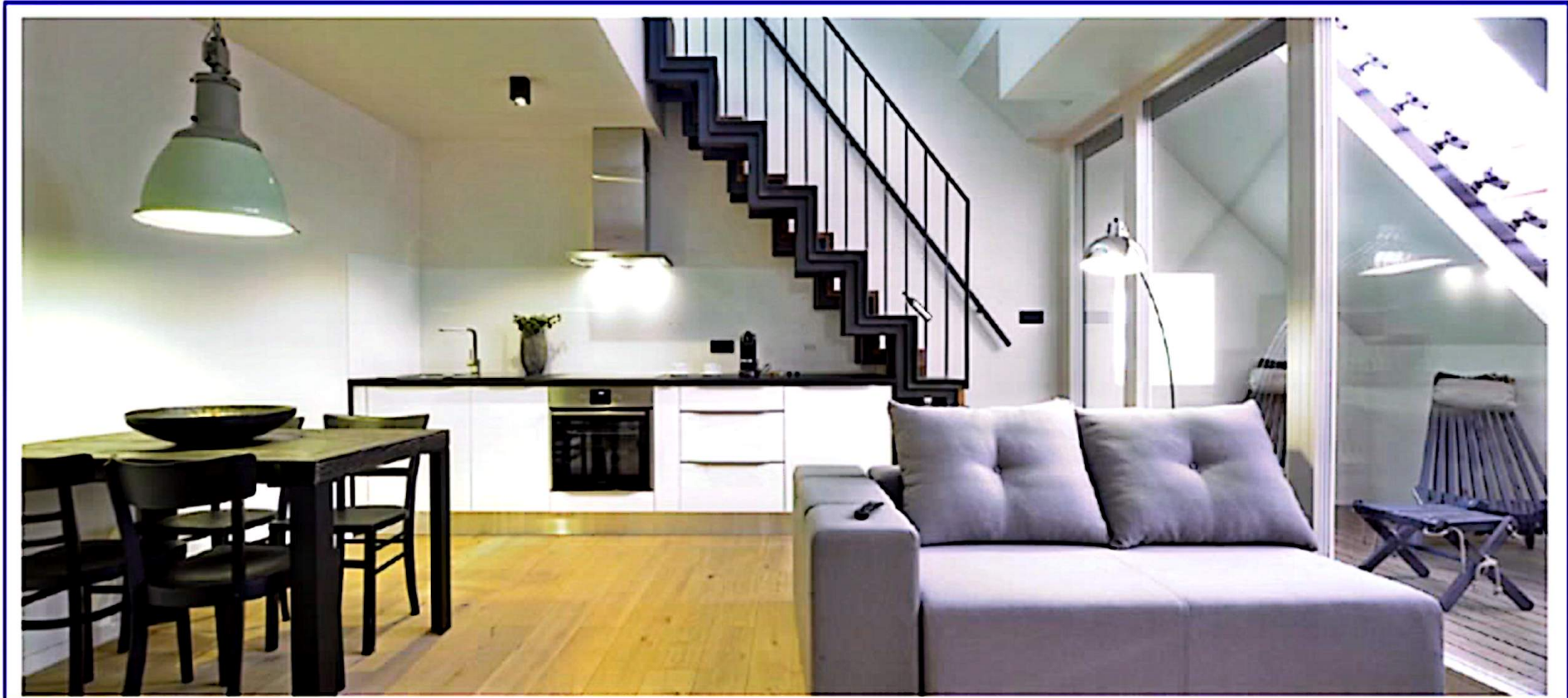
YOLO – The Input Image



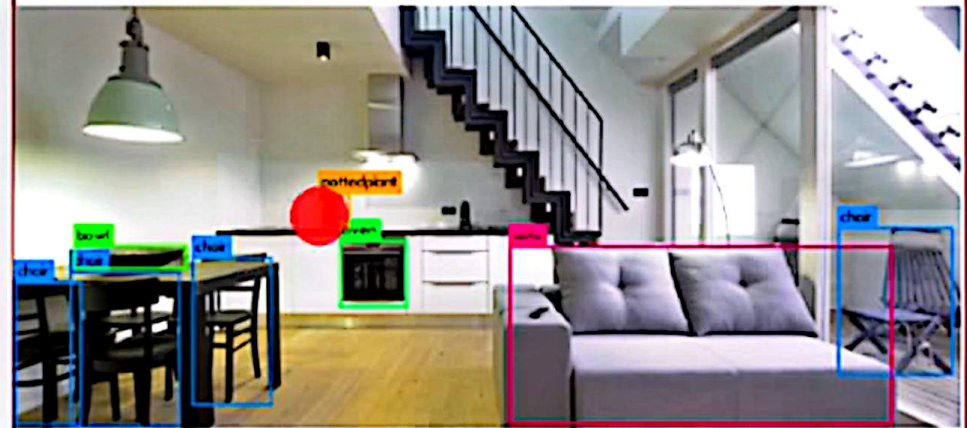
YOLO – The Detected Output



YOLO – The Input Image



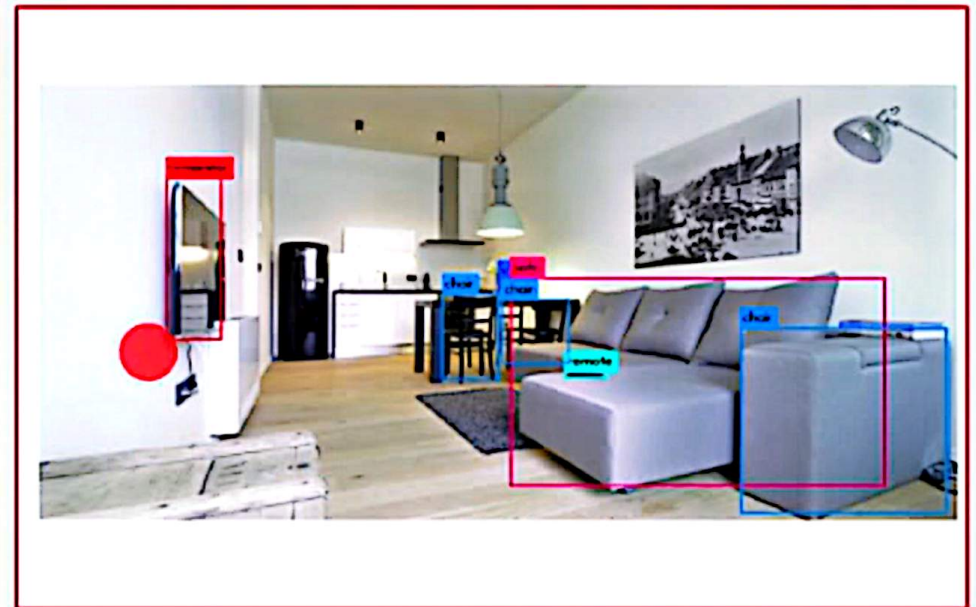
YOLO – The Detected Output



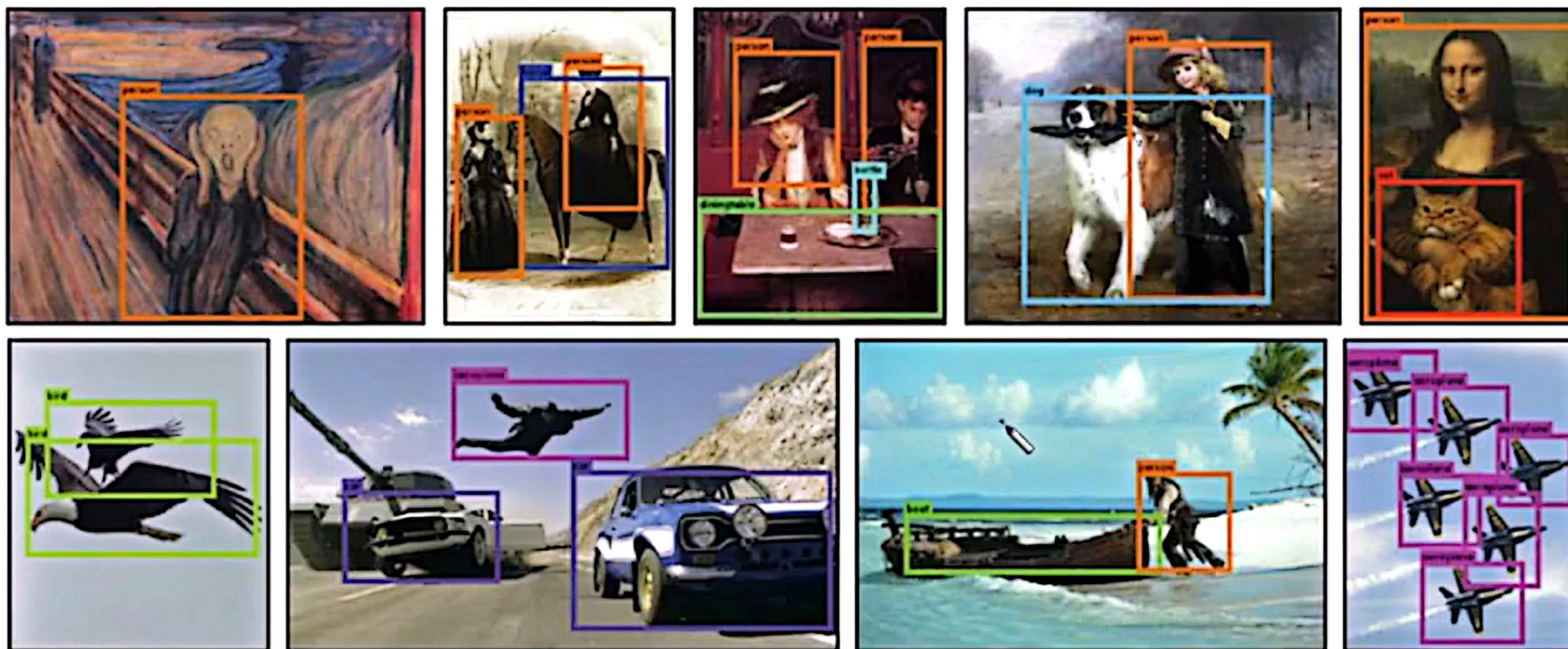
YOLO – The Input Image



YOLO – The Detected Output



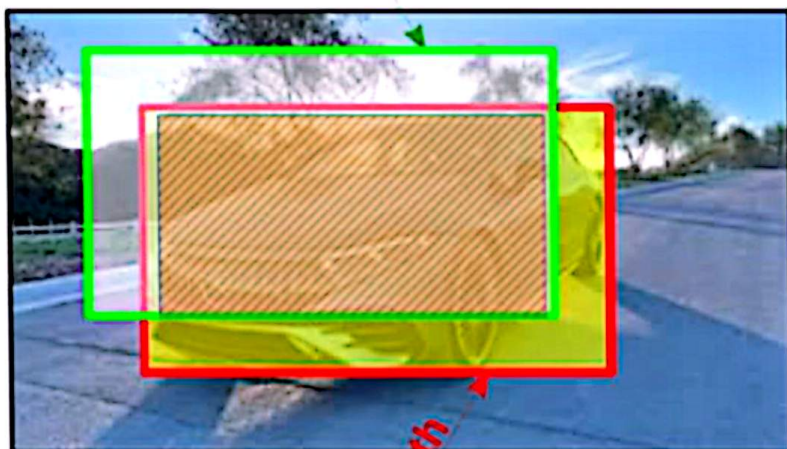
YOLO – Detection of Artificial Images



Intersection over Union

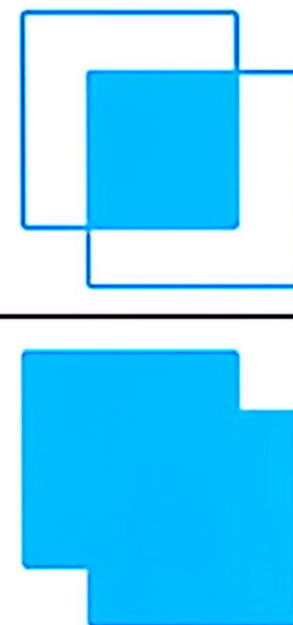
Intersection Over Union (IoU) (Evaluation of Detection Results Accuracy)

Detected



Ground Truth

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



Acceptance criteria $\text{IOU} > 0.6$ (The higher the better)

IOU is a measure of overlap between two bounding boxes

IOU Loss function

The Higher the IOU, the Better the performance

It is required to **Maximize** the **IOU** of the model output

It is required to **Minimize** (1-IOU) of the model output

Minimize the IOU Loss using Gradient Descent

$$Loss_{IOU} = 1 - IOU$$

Variants of IoU

- **GloU** (Generalized IoU)
- **DloU** (Distance IoU)
- **CloU** (Complete IoU)
- **EloU** (Efficient IoU)
- **Focal-EIoU**
- **SloU** (SCYLLA-IoU)
- **WIoU** (Wise-IoU)

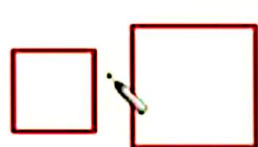
Intersection Over Union (IoU) in YOLO

- [1] Used as **Quality measure** of Predicted Bounding Boxes
- [2] Used **in Loss Calculation** (YOLO Version 2,3,)
- [3] Used in **Assignment of Anchor Box to Object** Bounding Box
- [4] Used in Selection of BEST Predicted Box (**in Non-Max Suppression** Algorithm)
- [5] Used as Similarity Measure in K-Means Clustering Algorithm (in order to **select Best “K” Anchors** sizes and Aspect Ratios)

Different Version of YOLO use different variants of IOU

You Only Look Once
YOLO

Review: Sliding Window Concept



Predefined Sliding Windows Templates

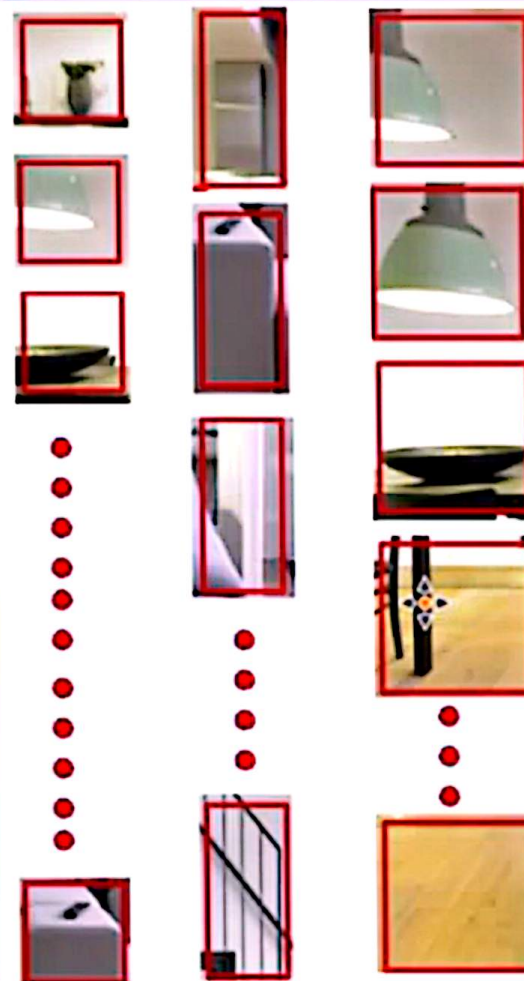


Image is Scanned by Multiple Windows with Different Sizes.
Sub Images are extracted and annotated with:
(W,H) of Sliding window and (x,y) of sliding window position

System is trained to classify each sub image
(No training for W,H,x,y)

The more windows sizes, the more accurate W,H of object
The more smooth sliding, the more accurate Position of object

Sub Images Captured by Sliding Windows



Prediction

$P_{c1}, P_{c2}, P_{c3}, P_{c4}, \dots$

$P_{c1}, P_{c2}, P_{c3}, P_{c4}, \dots$

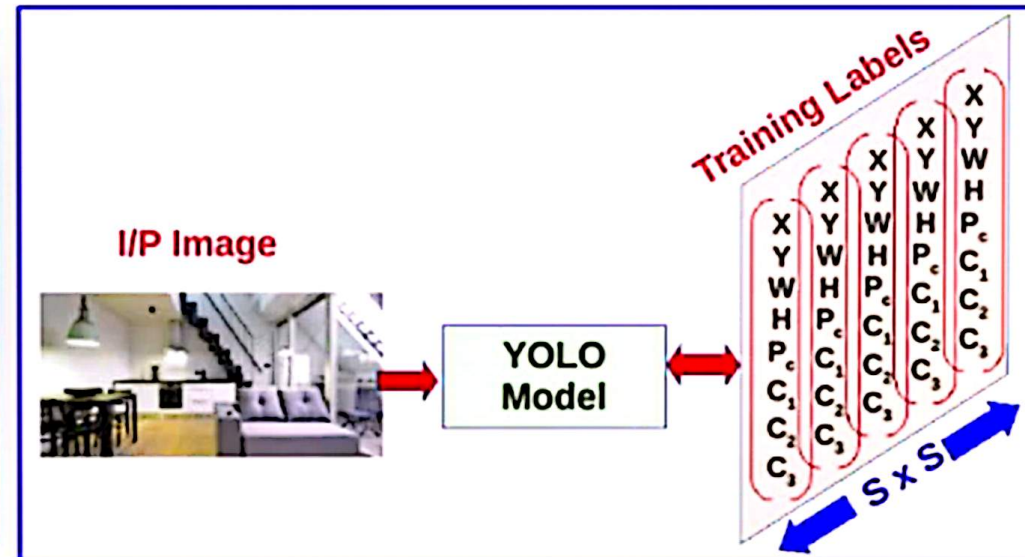
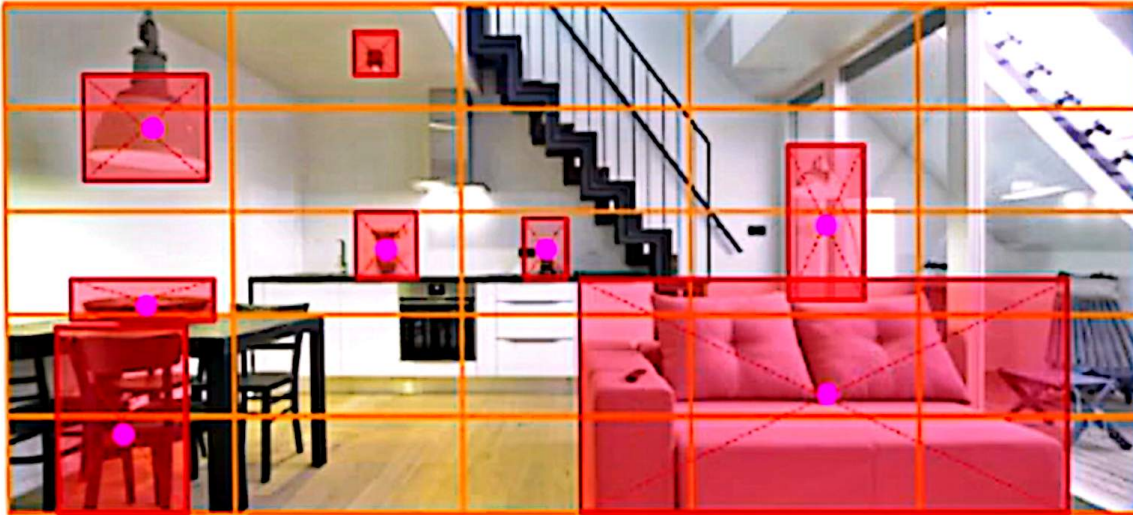
$P_{c1}, P_{c2}, P_{c3}, P_{c4}, \dots$

$P_{c1}, P_{c2}, P_{c3}, P_{c4}, \dots$

$P_{c1}, P_{c2}, P_{c3}, P_{c4}, \dots$

YOLO Concept at a Glance

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- Divide Image into GRID of **SxS Cells**.
- Annotate All objects with bounding Box and define Object Class.
- Define the CENTER of Each Bounding Box,
- Associate each bounding box with CELL which contains its Center

Construct Y vector (Label) for each CELL which contain:

Coord (x,y,W,H) of bounding box,
Class of the Object (One Hot Encoded),
Confidence value that there is an object center in this cell

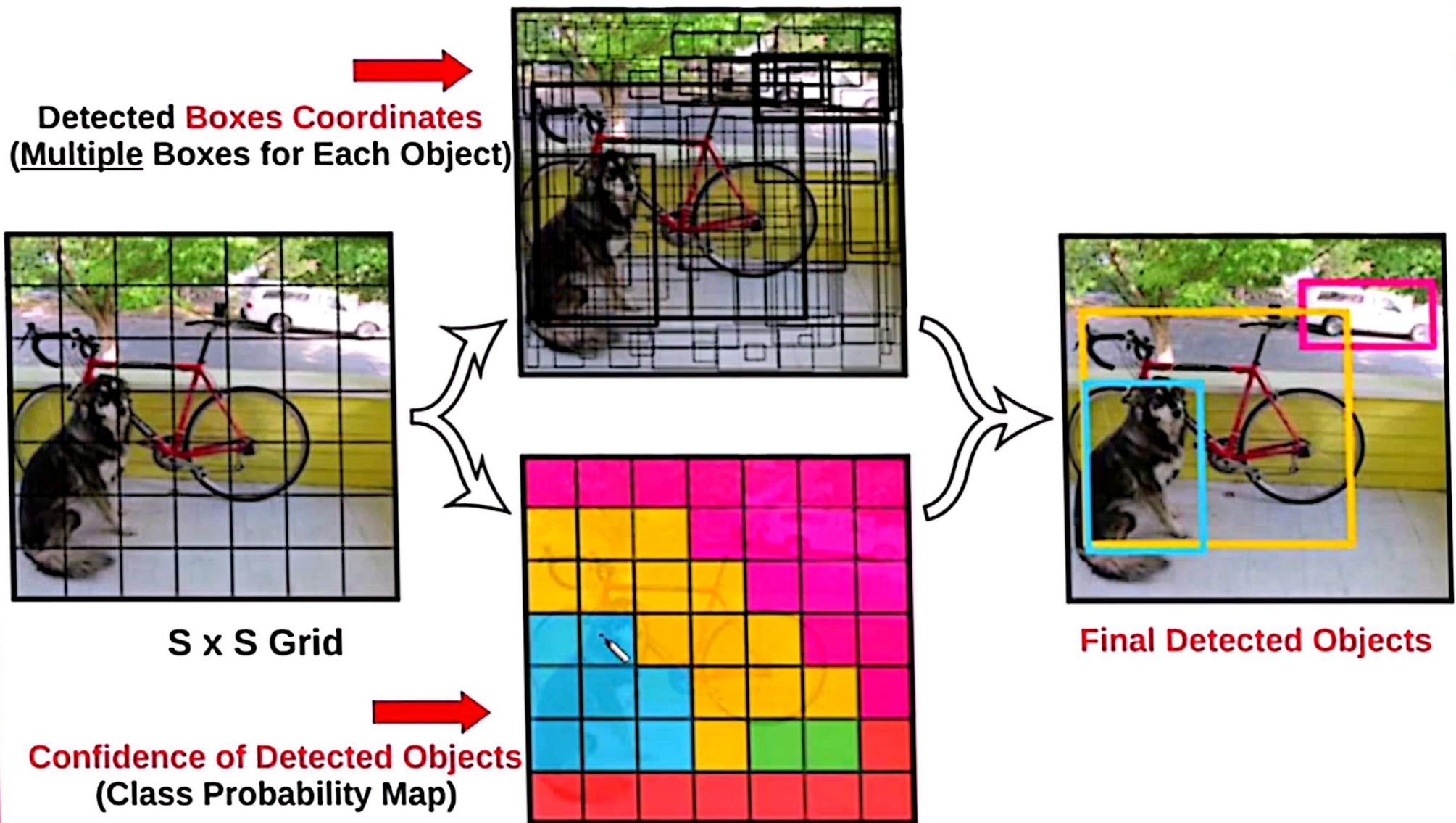
YOLO Model is trained to Predict:

For Each Cell

- Is there an object in this cell (P_c)
- Exact Location of Object (x,y)
- Exact Size of Object (W,H)
- Object Class ($C_1, C_2, \dots$) [same as Sliding Window]

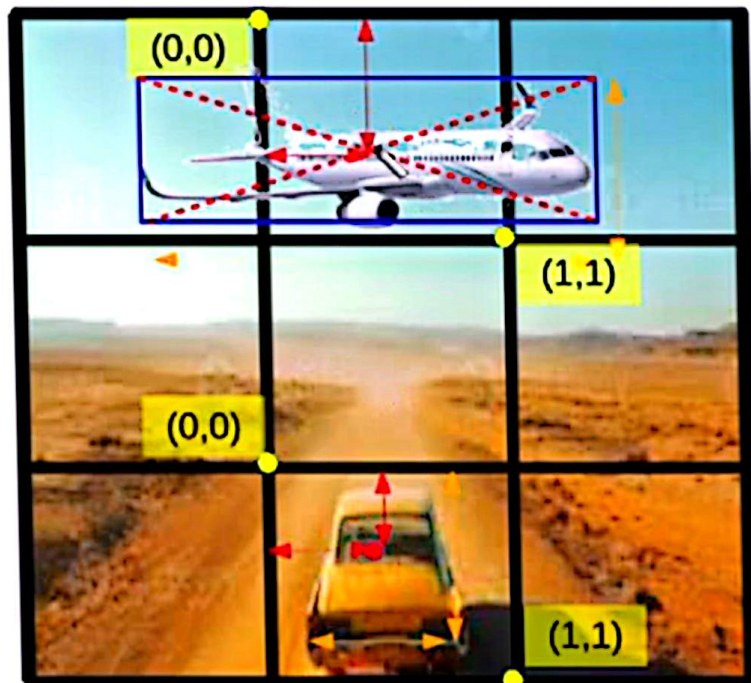
Problems:

Same Object may be Predicted in multiple Cells ??
Cell Contains Multiple Objects ??



Encoding of Object Coordinate ([1] Cell Based Encoding)

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X, Y, W, H
Plane: 0.4, 0.7, 1.9, 0.6
Car: 0.5, 0.4, 0.5, 0.7

Object Center

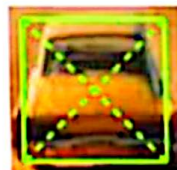


Image is divided into Grid Cells.

Each Grid box is associated with ONE Object Center (One Anchor)

For Each Grid Cell,

(0,0) is the top left corner

(1,1) is the bottom right corner

X,Y are relative to top left corner of **Cell**

X,Y values range from (0 to 1)

Width and height may exceed "1"

Encoding of Object Coordinate ([2] Image Based Encoding)

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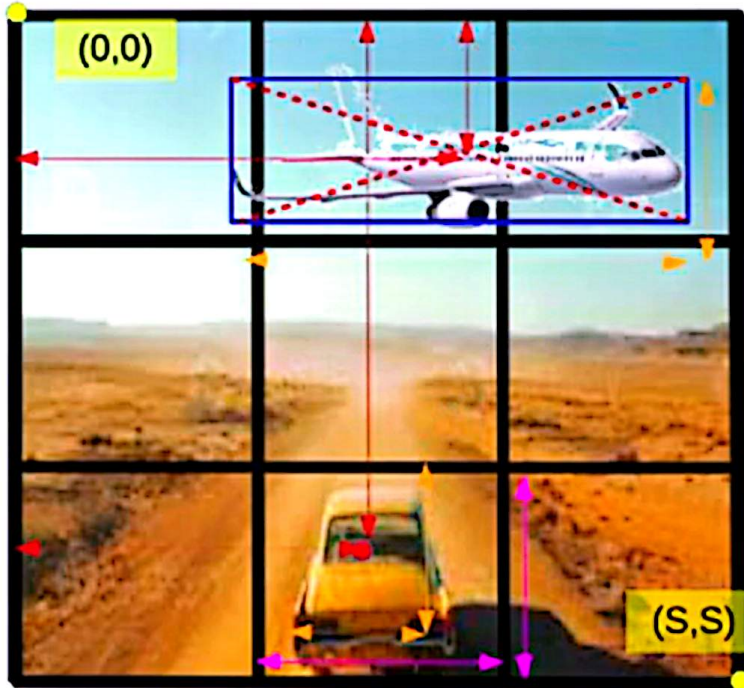


Image is divided into $S \times S$ Cells
(e.g. 3x3 Cells)

	X	Y	W	H
Plane:	1.8	0.7	1.9	0.6
Car:	1.5	2.4	0.5	0.7

Image is divided into Grid cells.

**Each Grid box is associated with
ONE Object Center (One Anchor)**

Using Cells Unit

$\longleftrightarrow = 1$

(0,0) is the top left corner of the IMAGE

(S,S) is the bottom right corner of the Image

X,Y are relative to top left corner of Image

X,Y values range from (0.x to S)

Width and height range from (0.x to S)

Anchor Boxes

Anchor Boxes (For Multiple Objects Detection)

One Anchor Box

Remember: One Grid Cell Contains One Object at Most (No Overlapped Objects)

Max Number of Detected Objects = Cell Count

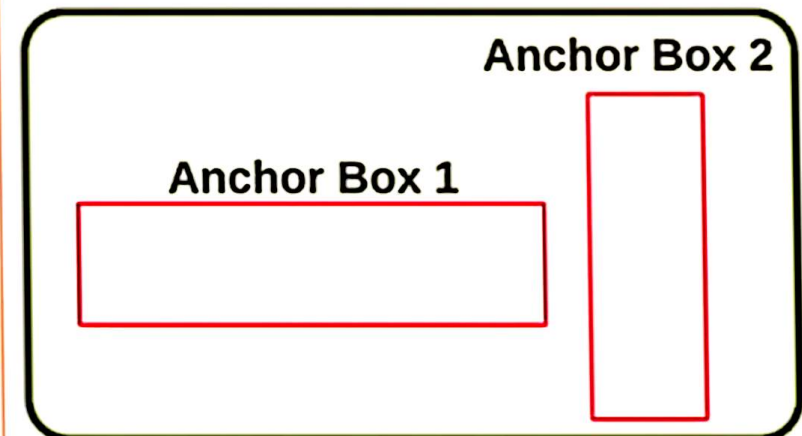
Each Object in training image is assigned to grid cell that contains that object's midpoint

What if there are multiple objects in a single grid?

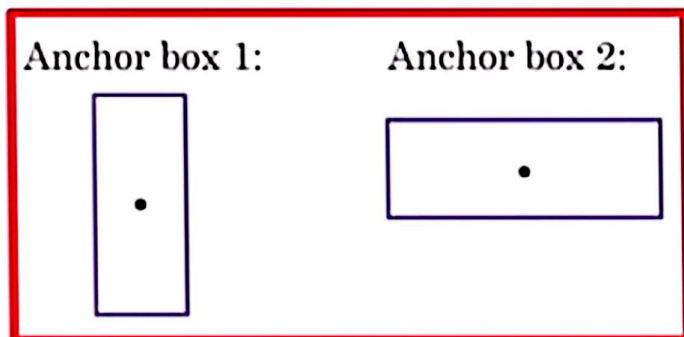
That can so often be the case in reality in Many Cases Such as:

- Overlapped Objects in Same grid
- Large grid size and small objects

That leads to the concept of anchor boxes



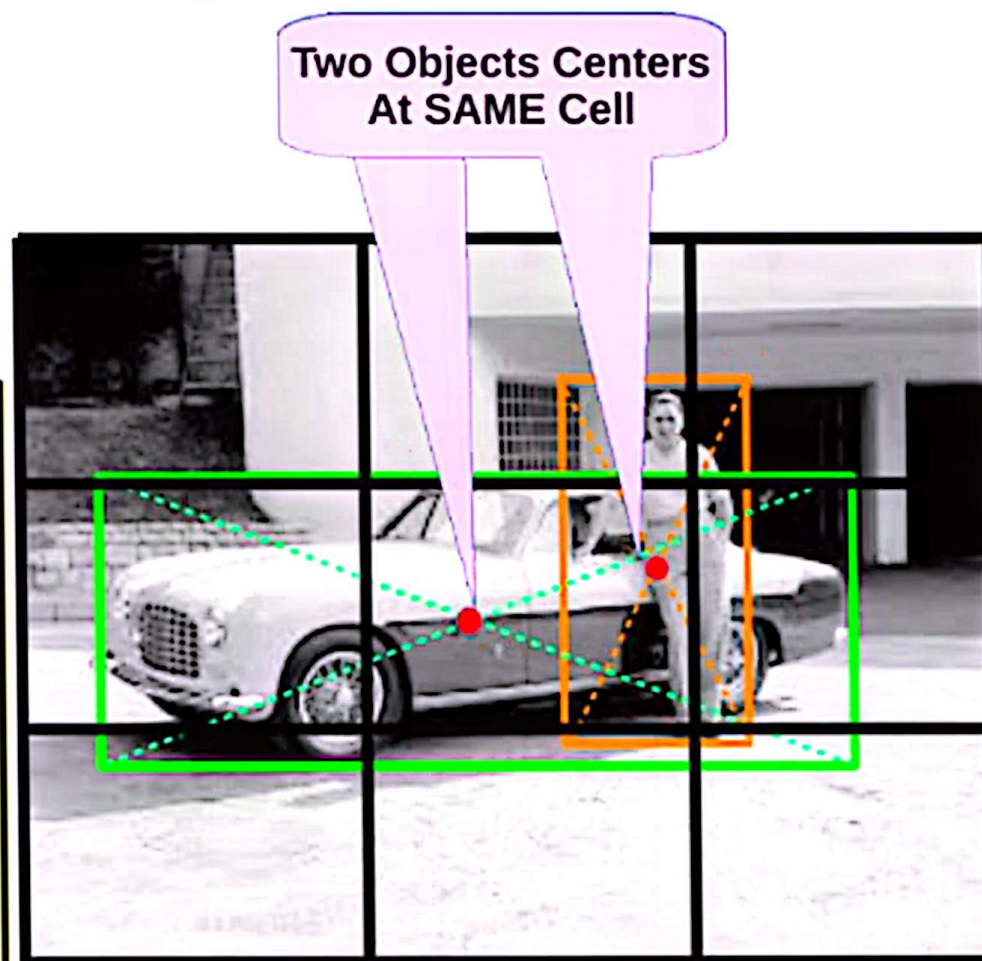
Multiple Anchor Boxes (for Overlapped Objects)



The objects are assigned to the anchor boxes based on:

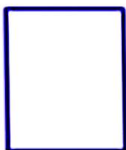
The similarity of the bounding boxes and the anchor box shape.

Since the shape of anchor box 1 is similar to the bounding box for the person, the **Person** will be assigned to **anchor box 1** and the **car** will be assigned to **anchor box 2**.



Multiple Anchor Boxes (for Small Objects)

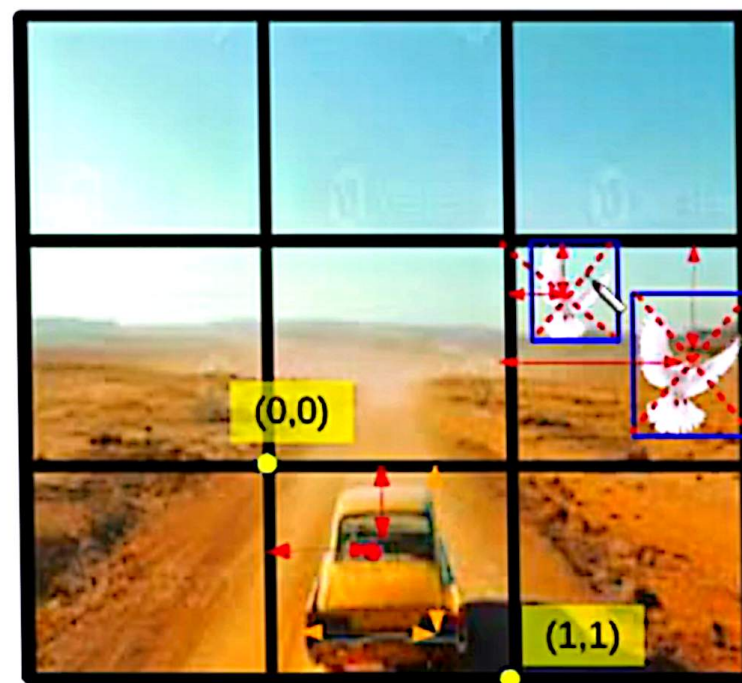
Anchor box 1:

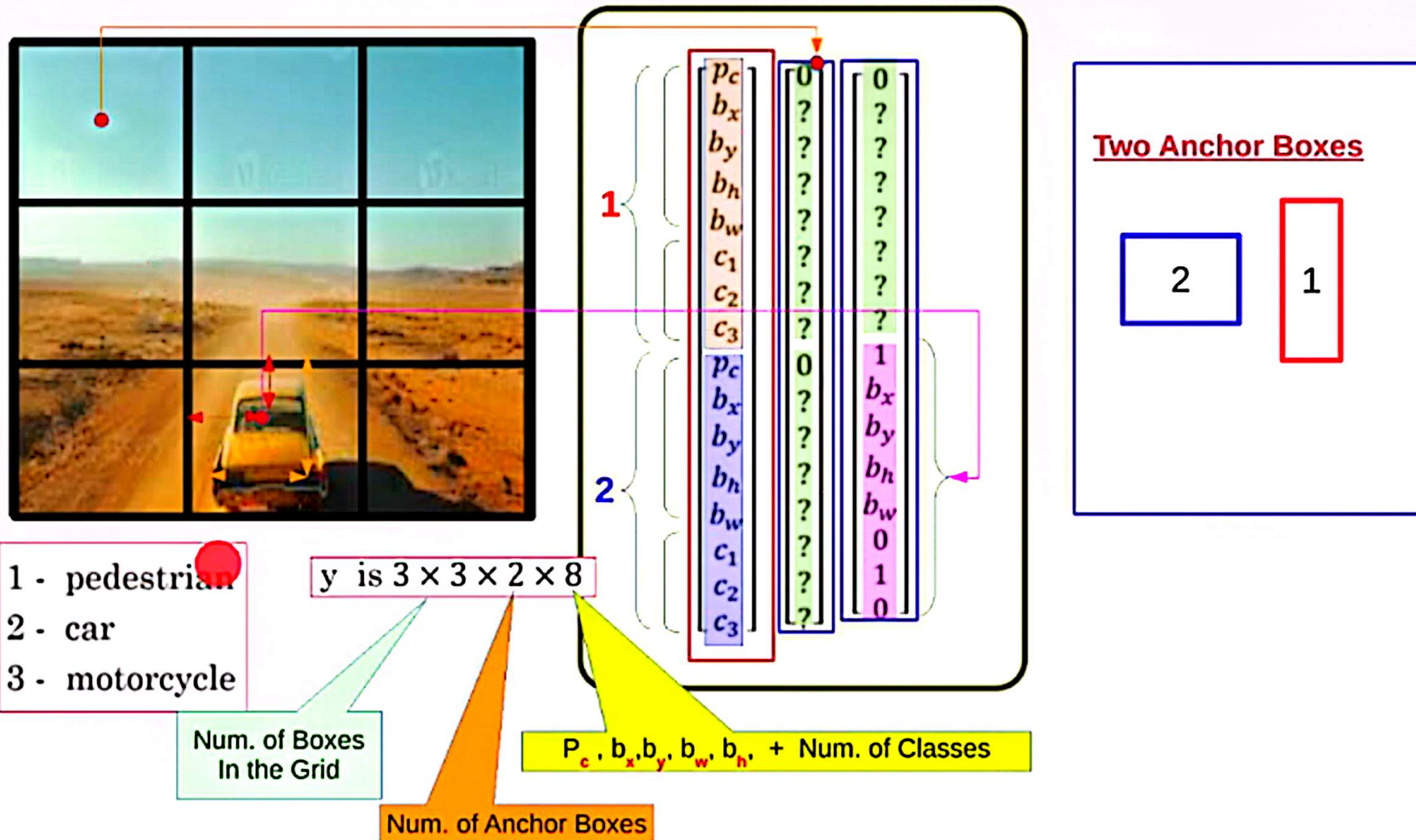


Anchor box 2:



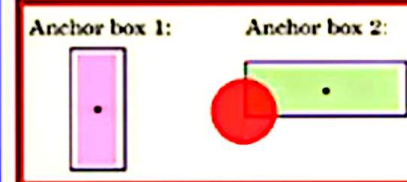
- Select Anchor Box Similar to Each Object (size and Aspect Ratio)
- Each Anchor Box will be associated with (Object X,Y,W,H, Object Class Type, In Addition to Confidence Probability that there Exist an object)





What if Object is Assigned to Wrong Anchor Box?

P_c	1	1	1	1	0	0
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	0.4	0.5	0.6	2.8	...	...
H	3.1	2.3	2.9	.25	...	...
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...
P_c	0	0	0	0	1	1
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	...	...	...	...	2.5	3.4
H	...	...	...	...	0.3	.35
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...

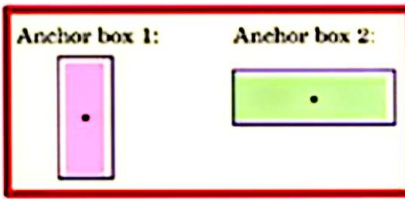


Wrong Anchor

Training Data is **NOT** Consistent $\rightarrow$ Less Accuracy

What if Object is Assigned to Wrong Anchor Box?

P_c	1	1	1	0	0	0
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	0.4	0.5	0.6	...	...	...
H	3.1	2.3	2.9	...	...	...
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...
P_c	0	0	0	1	1	1
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	...	...	...	2.8	2.5	3.4
H	...	...	...	.25	0.3	.35
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...

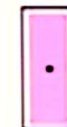


Proper Anchor Training Data is Consistent → High Accuracy

What if Object is Assigned to Wrong Anchor Box?

P_c	1	1	1	0	0	0
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	0.4	0.5	0.6	...	...	...
H	3.1	2.3	2.9	...	...	...
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...
P_c	0	0	0	1	1	1
X	...	...	...	...	...	...
Y	...	...	...	...	...	...
W	...	...	...	2.8	2.5	3.4
H	...	...	...	.25	0.3	.35
C_1	...	...	...	...	...	...
C_2	...	...	...	...	...	...
C_3	...	...	...	...	...	...

Anchor box 1:



Anchor box 2:



Proper Anchor

Training Data is Consistent → High Accuracy

Multiple Anchors, ONE Object / Cell

For $S \times S$ Cells, M Anchors, N Classes

One Object per Cell

$$\text{Total Size} = [S \times S] * [M * 5 + N]$$

Multiple Objects per Cell

$$\text{Total Size} = [S \times S] * M * [5 + N]$$

Example:

$S=13$, $M=6$ Anchors, $N=100$ Class

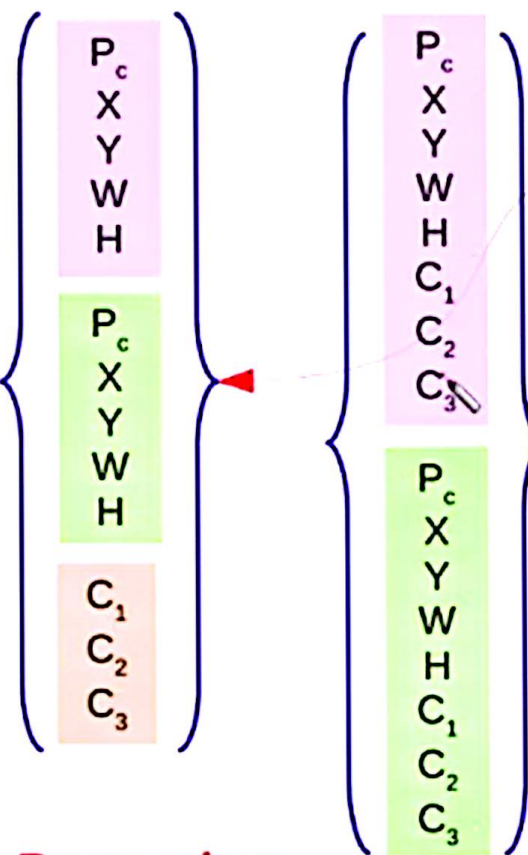
One Object per Cell

$$\text{Total Size} = 13 * 13 * [6 * 5 + 100] = 21970$$

Multiple Objects per Cell

$$\text{Total Size} = 13 * 13 * 6 * [5 + 100] = 106470$$

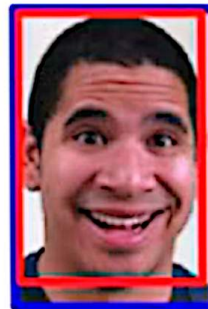
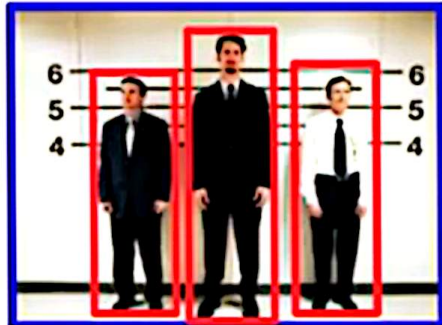
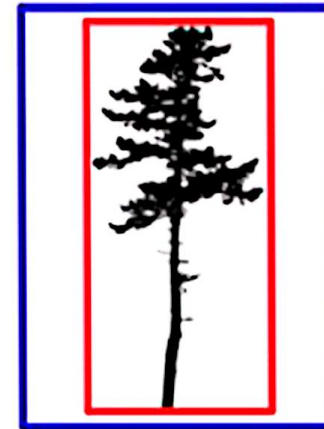
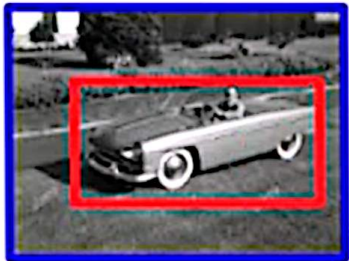
1 : 4.84



Remember:
ONE Vector / Cell

How to Define Sizes of Anchor Boxes

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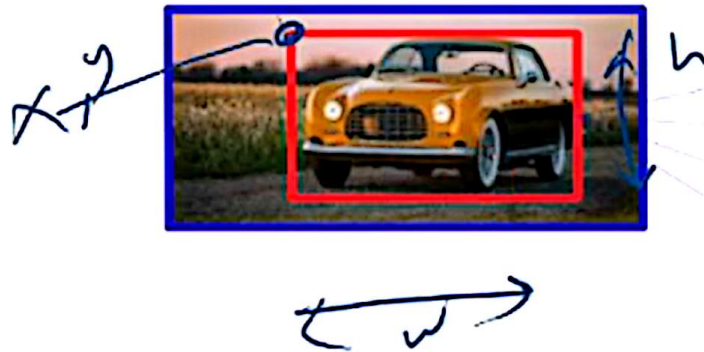
Objects with
Ground Truth



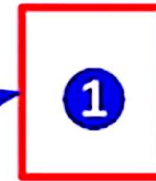
How to Assign Anchor Box to Object

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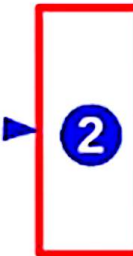
For Each Object Bounding Box



Calculate IOU_1



Calculate IOU_2



Calculate IOU_3



Calculate IOU_4



Anchor Boxes

Select Anchor Box with HIGHEST IOU

How to Define Number of Anchor Boxes

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K-Means Clustering

RUN with K= 3, 4, 5, 6, 7, 8, 9, Clusters]

Assign Ground Truth Bounding Box of Each Object to one of the "K" Anchor boxes (based on Max IOU)

Calculate IOU between Ground Truth Bounding Box of Each Object and assigned Anchor Box, Calculate Mean IOU

Repeat for K=2, 3, 4,, Obtain Mean IOU for each K

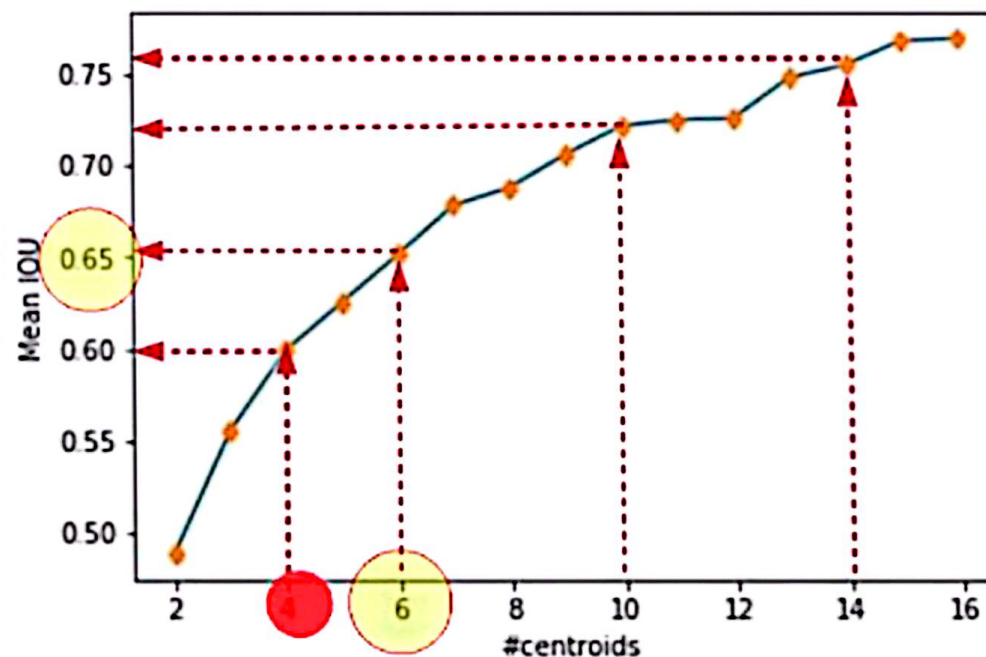
Select Minimum "K" Value that achieves "Acceptable IOU" [e.g. >0.6]

How to Define Number of Anchor Boxes

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K-Means Clustering

RUN with K= 3, 4, 5, 6, 7, 8, 9, Clusters]



Select Minimum “K” Value that achieves “**Acceptable IOU**” [e.g. >0.6]

YOLO Versions (till March 2025)

YOLO,	YOLO-NAS,
YOLOv2,	YOLO-World,
YOLOv3,	YOLOv9,
YOLOv4,	YOLOv10,
YOLOv5,	YOLOv11, and
YOLOv6,	YOLOv12.
YOLOv7,	
YOLOv8,	